12. Battery Backup Function

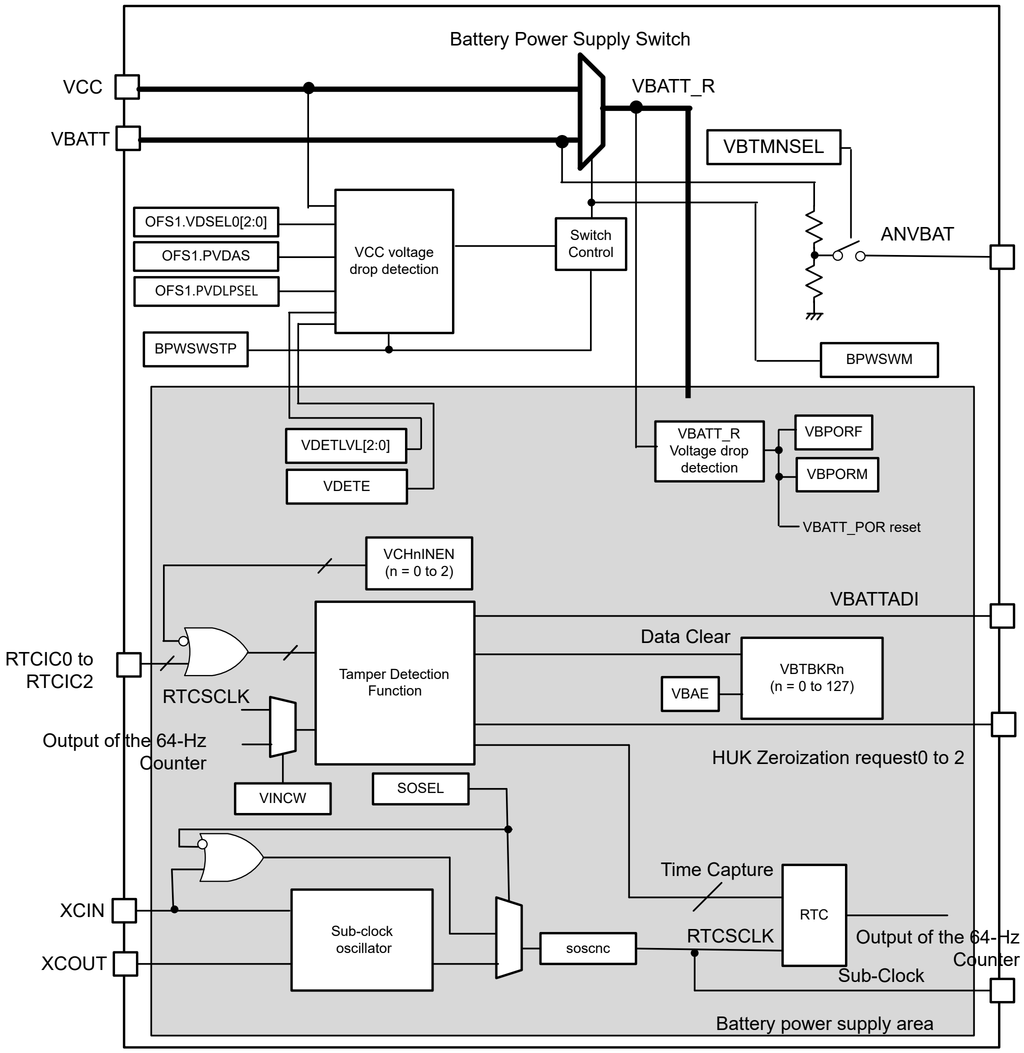
12.1 Overview

The MCU provides a battery backup function that maintains partial battery powering in the event of a power loss. Switching between VCC and VBATT, the battery-powered area includes RTC, SOSC, backup register, tamper detection and VBATT_R voltage drop detection. VBATT_R is the output voltage of the battery power supply switch. This is the power supply of the battery powered area.

During normal operation, the battery-powered area is powered by the main power supply, the VCC pin. When a VCC voltage drop is detected, the power source switches to the dedicated battery backup power pin, the VBATT pin. When the voltage rises again, the power source switches back from VBATT to VCC.

12.1.1 Battery Backup Function Block Diagram

Figure 12.1 shows the configuration of the battery backup function.



VBTMNSEL : VBATT Voltage Monitor Function Select Bit
VCHnINEN : VBATT CHn Input Enable bit (n = 0 to 2)
VDETLVL[2:0] : $V_{DETBATT}$ Level Select
VDETE : Voltage drop detection enable
BPWSWSTP : Battery Power Supply Switch Stop
VBPORF : VBATT_POR flag
BPWSWMON : Battery Power Supply Switch Status Monitor

VBTBKRn : VBATT Backup Register (n = 0 to 127)
VBAE : VBATT backup register access enable bit
OFS1 : Option Function Select Register 1
VDSEL0 : Voltage Detection 0 Level Select
PVDAS : Voltage Detection 0 Circuit Start
PVDLPSEL : PVD0 Select at Deep software standby mode
VINCW :VBATT Input Noise Canceler Width select
soscnc :Noise canceller for SOSC

Note: VCC : Main power supply pin
VBATT: Battery backup power supply pin
XCIN/XCOUT: SOSC input/output
RTCICn (n = 0 to 2): Input port for battery backup function

Figure 12.1 Battery backup function block diagram

12.1.2 Features of Battery Backup Function

The features include:

- Battery power supply switch
- VBATT_R voltage drop detection function
- Backup registers
- Time capture pin detection
- Tamper Detection function
- VBATT voltage monitor function

12.1.3 Battery Power Supply Switch

When the voltage applied to the VCC pin drops, this feature switches the power supply from the VCC pin to the VBATT pin. When the voltage rises, it switches the power supply from the VBATT pin back to the VCC pin.

When VCC is lower than $V_{DETBATT}$, and VCC is higher than VBAT, the injected current connects from the VCC to the VBATT pin through an internal diode. If the power supply battery connected to the VBATT pin cannot support this current injection, for example if the battery is not rechargeable, Renesas strongly recommends that you connect through a low-voltage threshold diode between the power supply battery and the VBATT pin.

It is necessary to enable voltage monitor 0 resets to use the battery backup function. The voltage monitor 0 level must be higher than the VBATT switch level.

12.1.4 VBATT_R Voltage Drop Detection Function

VBATT_R voltage drop detection function supports the battery-powered area. This function monitors the VBATT_R voltage level. VBATT_R is the output voltage of the battery power supply switch. This voltage drop detection causes a VBATT_POR reset and initializes the battery-powered area. See details in each register description. The VBATT status register includes a flag to check for this voltage drop detections.

12.1.5 Backup Registers

The battery-powered area provides 128-byte backup registers. These registers retain data when VBATT is supplied and VCC is powered off. When tampering is detected, data of backup register can be cleared to 0x00.

12.1.6 Time Capture Function

The RTC detects RTCICn (n = 0 to 2) pin input level change. For more information, see section 26, Realtime Clock (RTC).

12.1.7 Tamper Detection Function

The tamper detection function detects the RTCICn (n = 0 to 2) pin input event. The input event is defined as a change of RTCICn (n = 0 to 2) pin input level. The tamper detection flag is set to 1 by the input event. When interrupt is enabled and flag is set to 1, tamper detection interrupt is generated. When backup registers clear is enabled and flag is set to 1, the data of backup registers is cleared. Time Capture Function can select this flag as the source of the time capture trigger.

When Tamper Detection Zeroization Enable is enabled and flag is set to 1, the HUK Zeroization request is outputted.

12.1.8 VBATT Voltage Monitor Function

VBATT voltage monitor function is to monitor the input voltage level to the VBATT pin. The voltage level can be monitored as analog signal. For more information, see section 52, 16-bit A/D Converter (ADC16H).

12.2 Register Descriptions

12.2.1 BBFSAR : Battery Backup Function Security Attribute Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0x3D0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	NONSEC6	NONSEC5	NONSEC4	NONSEC3	NONSEC2	NONSEC1	NONSEC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	NONSEC0	Non-secure Attribute bit 0 Target register: VBATTMNSELR 0: Secure 1: Non-secure	R/W
1	NONSEC1	Non-secure Attribute bit 1 Target register: VBTBER 0: Secure 1: Non-secure	R/W
2	NONSEC2	Non-secure Attribute bit 2 Target register: VBTICTLR, VBTICTLR2, VBTIMONR 0: Secure 1: Non-secure	R/W
3	NONSEC3	Non-secure Attribute bit 3 Target register: VBTBPCR1, VBTBPCR2, VBTBPSR 0: Secure 1: Non-secure	R/W
4	NONSEC4	Non-secure Attribute bit 4 Target register: VBTADSR,VBTADCR1, VBTADCR2 0: Secure 1: Non-secure	R/W
5	NONSEC5	Non-secure Attribute bit 5 Target register: VBTADCR3 0: Secure 1: Non-secure	R/W
6	NONSEC6	Non-secure Attribute bit 6 Target register: VBTNCWCR 0: Secure 1: Non-secure	R/W
31:7	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1
Note: This register is write-protected by PRCR register.

The BBFSAR register controls the secure attribute of the battery backup function registers.

NONSEC0 bit (Non-secure Attribute bit 0)

This bit controls the security attribute of VBATTMNSELR.

NONSEC1 bit (Non-secure Attribute bit 1)

This bit controls the security attribute of VBTBER.

NONSEC2 bit (Non-secure Attribute bit 2)

This bit controls the security attribute of VBTICTLR, VBTICTLR2, and VBTIMONR.

NONSEC3 bit (Non-secure Attribute bit 3)

This bit controls the security attribute of VBTBPCR1 and VBTADCR2.

NONSEC4 bit (Non-secure Attribute bit 4)

This bit controls the security attribute of VBTADSR, VBTADCR1 and VBTADCR2.

NONSEC5 bit (Non-secure Attribute bit 5)

This bit controls the security attribute of VBTADCR3.

NONSEC6 bit (Non-secure Attribute bit 6)

This bit controls the security attribute of VBTNCWCR.

12.2.2 VBRSABAR : VBATT Backup Register Security Attribute Boundary Address Register

Base address: SYSC = 0x4001_E000

Offset address: 0x3B0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SABA[15:0]															
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	SABA[15:0]	Boundary address between secure and non-secure	R/W

Note: S-TYPE-6, P-TYPE-2

VBRSABAR specify the boundary address between Secure and non-secure regions of VBATT Backup Register. This register specifies lower 16 bits of VBATT Backup Register address. Secure region is less than SABA. Non-secure region is SABA or higher. The boundary address can set in units of 32 bytes, so the value written from b4 to b0 should be 0. SABA has no effect other than VBATT backup register.

VBATT Backup register is separated as follows.

Secure region : 0x4001_ED00 ≤ Address < 0x4001_0000 + SABA

Non-secure region : 0x4001_0000 + SABA ≤ Address

Note: The initial value specifies an address greater than the end address of VBATT Backup Register, so all area is Secure. If you specify all area of the VBATT Backup Register as Non-secure, specify the top address in the VBATT Backup Register area.

12.2.3 VBRPABARS : VBATT Backup Register Privilege Attribute Boundary Address Register for Secure Region

Base address: SYSC = 0x4001_E000

Offset address: 0x3B4

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PABAS[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PABAS[15:0]	Privilege Attribute Boundary Address for Secure Region Boundary address between Privileged area and any-Privileged area in Secure	R/W

Note: S-TYPE-6, P-TYPE-2

VBRPABARS specify the boundary address between Privileged area and Any-Privileged area in Secure regions of VBATT Backup Register. This register further separates the Secure region of VBATT Backup Register set by VBRSABAR register into Privileged and Any-Privileged areas. This register specifies lower 16 bits of VBATT Backup Register address. Privileged area is less than PABAS. Any-Privileged area is PABAS or higher. The boundary address can set in units of 32byte, so the value written from b4 to b0 should be 0. PABAS has no effect other than VBATT backup register.

VBATT Backup register is separated as follows.

Privileged area in Secure region : 0x4001_ED00 ≤ Address < 0x4001_0000 + PABAS
0x4001_ED00 ≤ Address < 0x4001_0000 + SABA

This area that satisfies both address conditions.

Any-Privileged area in Secure region : 0x4001_0000 + PABAS ≤ Address < 0x4001_0000 + SABA

Note: The initial value specifies before the top address of the VBATT Backup Register, so all the Secure region is Any-Privilege. In this case, the setting of this register is actually ignored.

If you specify Privileged for the entire VBATT Backup Register that you have set as the Secure region, specify VBRPABARS register to the 0xFFE0.

12.2.4 VBRPABARNS : VBATT Backup Register Privilege Attribute Boundary Address Register for Non-secure Region

Base address: SYSC_NS = 0x5001_E000

Offset address: 0x3B8

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PABANS[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PABANS[15:0]	Privilege Attribute Boundary Address for Non-secure Region Boundary address between Privileged area and Any-Privileged area in Non-secure	R/W

Note: S-TYPE-7, P-TYPE-2

VBRPABARNS specify the boundary address between Privileged area and Any-Privileged area in Non-secure regions of VBATT Backup Register. This register further separates the Non-secure region of VBATT Backup Register set by VBRSABAR register into Privileged and Any-Privileged areas. This register specifies lower 16 bits of VBATT Backup Register address. Privileged area is less than PABANS. Any-Privileged area is PABANS or higher. The boundary address can set in units of 32byte, so the value written from b4 to b0 should be 0. PABANS has no effect other than VBATT backup register.

VBATT Backup register is separated as follows.

Privileged area in Non-Secure region : 0x4001_0000 + SABA ≤ Address < 0x4001_0000 + PABANS

Any-Privileged area in Non-Secure region : 0x4001_0000 + PABANS ≤ Address
0x4001_0000 + SABA ≤ Address

This area that satisfies both address conditions.

Note: The initial value specifies before the top address of VBATT Backup Register, so all the Non-secure region is Any-Privilege. In this case, the setting of this register is actually ignored.

If you specify Privileged for the entire VBATT Backup Register that you have set as the Non-secure region, specify VBRPABARNS register to the 0xFFE0.

12.2.5 VBATTMNSELR : Battery Backup Voltage Monitor Function Select Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA84

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	VBTMNSEL
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VBTMNSEL	VBATT Voltage Monitor Function Select Bit Select VBATT voltage monitor function 0: Disables VBATT voltage monitor function 1: Enables VBATT voltage monitor function	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

VBATTMNSELR is the register which controls VBATT voltage monitor function.

This register is initialized by all reset sources except VBATT_POR reset.

VBTMNSEL bit (VBATT Voltage Monitor Function Select Bit)

Select VBATT low voltage monitor function. After setting this bit to 1, it is necessary to wait t_{MONWT} for the monitor level to stabilize.

Consumption current increases while VBTMNSEL = 1. So, after monitoring the VBATT voltage level, clear VBTMNSEL to 0 in order to reduce power consumption of VBATT power supply.

For more information on t_{MONWT}, see section 60, Electrical Characteristics.

12.2.6 VBTBER : VBATT Backup Enable Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC40

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	VBAE	—	—	—
Value after reset:	0	0	0	0	1	0	0	0

Bit	Symbol	Function	R/W
2:0	—	These bits are read as 0. The write value should be 0.	R/W
3	VBAE	VBATT backup register access enable bit 0: Disable to access VBTBKR 1: Enable to access VBTBKR	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

VBAE bit is initialized by the power on reset of VCC or after transition to VBATT mode.

VBAE bit (VBATT backup register access enable bit)

You must write 1 to VBAE before accessing VBTBKR. And you must write 0 after finishing all access (Write or Read) to VBTBKR. If you do not write 0 to VBAE, the data of VBTBKR does not be kept at VBATT mode.

Also, while VBAE is set to 0, the value of VBTBKR can be retained even if the VDD power supply and VCC power supply are powered off.

(The value can be retained even if the Core Voltage Monitor reset and the Overcurrent protection reset occurs after setting VBAE to 0.)

To access VBTBKR, wait for at least 500 ns after writing 1 to VBAE, and then access VBTBKR.

Before entering the Deep Software Standby mode, you must write 0 to VBAE.

To enter the deep software standby mode, wait at least 250 ns after writing 0 to VBAE, and then enter the Deep Software Standby mode.

If you don't use VBTBKR, you should change VBAE to 0 in order to reduce power consumption of VBTBKR.

12.2.7 VBTBKRn : VBATT Backup Register (n = 0 to 127)

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xD00 + 0x001 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	VBTBKRn[7:0]							
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	VBTBKRn[7:0]	VBATT Backup Register The value of this register is retained even in VBATT mode. This register is not initialized by any reset sources.	R/W

Note: S-TYPE-3, P-TYPE-3

VBTBKRn[7:0] bits (VBATT Backup Register)

The value of this register is retained even in VBATT mode.

You can use 32, 16 or 8-bit access instruction when accessing the VBTBKRn register. When accessing, please note that the byte order of the data stored in the VBTBKRn register is little endian.

The data of VBTBKRn register is cleared to 0x00 by VBATT_POR reset or tamper detection function.

12.2.8 VBTICTLR : VBATT Input Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC4C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	VCH2INEN	VCH1INEN	VCH0INEN
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VCH0INEN	VBATT CH0 Input Enable 0: RTCIC0 input disable 1: RTCIC0 input enable	R/W
1	VCH1INEN	VBATT CH1 Input Enable 0: RTCIC1 input disable 1: RTCIC1 input enable	R/W
2	VCH2INEN	VBATT CH2 Input Enable 0: RTCIC2 input disable 1: RTCIC2 input enable	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

VBTICTLR is the register that can select RTCICn (n = 0 to 2) pin as input. This register is only initialized by VBATT_POR reset.

VCHnINEN bits (VBATT CHn Input Enable Bits) (n = 0 to 2)

The VCHnINEN bit enables the input direction on the RTCICn. A wait time of 50 µs is required for the operation to stabilize after the RTCICn pin input is enabled.

12.2.9 VBTICTLR2 : VBATT Input Control Register 2

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC4D

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	VCH2 EG	VCH1 EG	VCH0 EG	—	VCH2 NCE	VCH1 NCE	VCH0 NCE
Value after reset:	0	1	1	1	0	0	0	0

Bit	Symbol	Function	R/W
0	VCH0NCE	VBATT CH0 Input Noise Canceler Enable 0: RTCIC0 pin input noise canceler disable 1: RTCIC0 pin input noise canceler enable	R/W
1	VCH1NCE	VBATT CH1 Input Noise Canceler Enable 0: RTCIC1 pin input noise canceler disable 1: RTCIC1 pin input noise canceler enable	R/W
2	VCH2NCE	VBATT CH2 Input Noise Canceler Enable 0: RTCIC2 pin input noise canceler disable 1: RTCIC2 pin input noise canceler enable	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	VCH0EG	VBATT CH0 Input Edge Select 0: RTCIC0 pin input event is detected on falling edge 1: RTCIC0 pin input event is detected on rising edge	R/W
5	VCH1EG	VBATT CH1 Input Edge Select 0: RTCIC1 pin input event is detected on falling edge 1: RTCIC1 pin input event is detected on rising edge	R/W
6	VCH2EG	VBATT CH2 Input Edge Select 0: RTCIC2 pin input event is detected on falling edge 1: RTCIC2 pin input event is detected on rising edge	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

VBTICTLR2 is the register that can select RTCICn (n = 0 to 2) pin input mode. This register is only initialized by VBATT_POR reset.

VCHnNCE bits (VBATT CHn Input Noise Canceler Enable) (n = 0 to 2)

The VCHnNCEN bit enables the input noise canceler on the RTCICn (n = 0 to 2) pin input.

VCHnEG bits (VBATT CHn Input Edge Select) (n = 0 to 2)

The VCHnEG bit selects input event detection edge to set VBTADFn flag.

12.2.10 VBTIMONR : VBATT Input Monitor Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC4E

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	VCH2 MON	VCH1 MON	VCH0 MON
Value after reset:	0	0	0	0	0	x	x	x

Bit	Symbol	Function	R/W
0	VCH0MON	VBATT CH0 Input monitor 0: RTCIC0 pin input is low level 1: RTCIC0 pin input is high level.	R
1	VCH1MON	VBATT CH1 Input monitor 0: RTCIC1 pin input is low level 1: RTCIC1 pin input is high level	R
2	VCH2MON	VBATT CH2 Input monitor 0: RTCIC2 pin input is low level 1: RTCIC2 pin input is high level.	R
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

VBTIMONR is the register that can monitor RTCICn (n = 0 to 2) pin input level.

VCHnMON bits (VBATT CHn Monitor) (n = 0 to 2)

The VCHnMON bit indicates input level on the RTCICn (n = 0 to 2) pin.

12.2.11 VBTBPCR1 : VBATT Battery Power Supply Control Register 1

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xA88

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	BPWSWSTP
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BPWSWSTP	Battery Power Supply Switch Stop 0: Battery power supply switch enable 1: Battery power supply switch stop	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

The VBTBPCR1 register controls battery power supply switch. This register is initialized by all reset sources except Deep Software Standby reset and VBATT_POR reset.

BPWSWSTP bit (Battery Power Supply Switch Stop)

The BPWSWSTP bit can enable switching the backup module supply power source from VCC to VBATT when the voltage applied to the VCC pin drops.

When stop is selected, the battery backup module power supply is always from VCC. Set the VDETE bit to 0 after setting theBPWSWSTP bit to 1. The BPWSWSTP bit must not be set from 1 to 0 while the VDETE bit is 1. If this setting is made, the state of the backup power area cannot be guaranteed.

If the BPWSWSTP bit is changed from 1 to 0, set the value of the VBTBPCR2.VDETLVL before it.

After setting the BPWSWSTP value from 1 to 0, wait for t_{DETWT} then set it to the VBTBPCR2.VDETE bit is 1.

12.2.12 VBTBPCR2 : VBATT Battery Power Supply Control Register 2

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC45

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	VDETE	—	VDETLVL[2:0]		
Value after reset:	0	0	0	0	0	1	1	0

Bit	Symbol	Function	R/W
2:0	VDETLVL[2:0]	$V_{DETBATT}$ Level Select 0 0 0: 2.80 V 0 0 1: 2.53 V 0 1 0: 2.10 V 0 1 1: 1.95 V 1 0 0: 1.85 V 1 0 1: 1.75 V 1 1 0: setting prohibited 1 1 1: setting prohibited	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	VDETE	Voltage drop detection enable 0: VCC Voltage drop detection disable 1: VCC Voltage drop detection enable	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

The VBTBPCR2 register controls battery power supply switch. This register is only initialized by VBATT_POR reset.

VDETLVL[2:0] bits ($V_{DETBATT}$ Level Select)

The VDETLVL[2:0] bits select $V_{DETBATT}$ level. When the voltage of the VCC pin drops below $V_{DETBATT}$, the power supply source switches from the VCC pin to the VBATT pin. The $V_{DETBATT}$ level should be below Voltage monitoring 0 level.

This MCU can reduce power consumption of the Deep Software Standby mode 1 or 2. When the MCU enters the Deep Software Standby mode 1 or 2 with that OFS1.PVDAS = 0 and OFS1.PVDLPSEL = 0, the battery power supply switch is controlled by the voltage monitor 0 which has lower power consumption than VCC voltage drop detection circuit dedicated for VBATT.

The VDETLVL[2:0] bits should change while the VDETE bit is 0. The VCC voltage detection function needs t_{DETWT} wait time for stabilization at the change of $V_{DETBATT}$ level. The VDETE bit should be set to 1 after this wait time.

For details on VDETLVL[2:0] bits, see section 12.3.2. VBATT Battery Power Supply Switch Usage.

VDETE bit (Voltage drop detection enable)

The VDETE bit enables the VCC voltage drop detection function for the battery power supply switch. The initial value of the VDETE bit is 0, at this time, the VCC voltage drop detection function for the battery power supply switch is disabled. If you use the battery power supply switch function, you must select $V_{DETBATT}$ level and enables the VCC voltage drop detection function.

For details on VDETE bit, see section 12.3.2. VBATT Battery Power Supply Switch Usage.

For more information on t_{DETWT}, see section 60, Electrical Characteristics.

12.2.13 VBTBPSR : VBATT Battery Power Supply Status Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC46

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	BPWSWM	VBPORM	—	—	—	VBPORF
Value after reset:	0	0	x	x	0	0	0	x

Bit	Symbol	Function	R/W
0	VBPORF	VBATT_POR Flag 0: VBATT_R voltage drop is not detected 1: VBATT_R voltage drop is detected	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	VBPORM	VBATT_POR Monitor 0: VBATT_R voltage < $V_{PORBATT}$ 1: VBATT_R voltage > $V_{PORBATT}$	R
5	BPWSWM	Battery Power Supply Switch Status Monitor 0: VCC voltage < $V_{DETBATT}$ 1: VCC voltage > $V_{DETBATT}$	R
7:6	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

The VBTBPSR register indicate battery power supply status.

VBPORF flag (VBATT_POR Flag)

The VBPORF flag indicates that VBATT_R voltage drop was detected and VBATT_POR reset was asserted.

[Setting condition]

- When VBATT_R voltage drops below V_{PORBAT}.

[Clearing condition]

- When 0 is written.

VBPORM bit (VBATT_POR Monitor)

The VBPORM bit indicates comparison result between VBATT_R and V_{PORBAT}.

BPWSWM bit (Battery Power Supply Switch Status Monitor)

The BPWSWM bit indicates comparison result between VCC and $V_{DETBATT}$.

12.2.14 VBTADSR : VBATT Tamper Detection Status Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC48

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	VBTADF2	VBTADF1	VBTADF0
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VBTADF0	VBATT Tamper Detection flag 0 0: RTCIC0 input edge is not detected 1: RTCIC0 input edge is detected	R/W
1	VBTADF1	VBATT Tamper Detection flag 1 0: RTCIC1 input edge is not detected 1: RTCIC1 input edge is detected	R/W
2	VBTADF2	VBATT Tamper Detection flag 2 0: RTCIC2 input edge is not detected 1: RTCIC2 input edge is detected	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

The VBTADSR register indicate tamper detection function status. This register is only initialized by VBATT_POR reset.

VBTADFn flags (VBATT Tamper Detection flag n) (n = 0, 1, 2)

The VBTADFn (n = 0, 1, 2) flag indicates that RTCICn (n = 0, 1, 2) input edge is detected. The edge type of RTCICn (n = 0, 1, 2) input can be selected by VCHnEG (n = 0, 1, 2) bit.

[Setting condition]

- When RTCICn (n = 0, 1, 2) input edge is detected.

[Clearing conditions]

- When 0 is written after 1 is read from VBTADFn (n = 0, 1, 2) flag.

12.2.15 VBTADCR1 : VBATT Tamper Detection Control Register 1

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC49

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	VBTADCE2	VBTADCE1	VBTADCE0	—	VBTADIE2	VBTADIE1	VBTADIE0
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VBTADIE0	VBATT Tamper Detection Interrupt Enable 0 0: Interrupt by VBTADF0 flag is disable 1: Interrupt by VBTADF0 flag is enable	R/W
1	VBTADIE1	VBATT Tamper Detection Interrupt Enable 1 0: Interrupt by VBTADF1 flag is disable 1: Interrupt by VBTADF1 flag is enable	R/W
2	VBTADIE2	VBATT Tamper Detection Interrupt Enable 2 0: Interrupt by VBTADF2 flag is disable 1: Interrupt by VBTADF2 flag is enable	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	VBTADCE0	VBATT Tamper Detection Backup Register Clear Enable 0 0: Clear Backup Register by VBTADF0 flag is disable 1: Clear Backup Register by VBTADF0 flag is enable	R/W
5	VBTADCE1	VBATT Tamper Detection Backup Register Clear Enable 1 0: Clear Backup Register by VBTADF1 flag is disable 1: Clear Backup Register by VBTADF1 flag is enable	R/W
6	VBTADCE2	VBATT Tamper Detection Backup Register Clear Enable 2 0: Clear Backup Register by VBTADF2 flag is disable 1: Clear Backup Register by VBTADF2 flag is enable	R/W

Bit	Symbol	Function	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

The VBTADCR1 register control tamper detection functions. This register is only initialized by VBATT_POR reset.

VBTADIEn bits (VBATT Tamper Detection Interrupt Enable) (n = 0, 1, 2)

The VBTADIEn (n = 0, 1, 2) bit enables Tamper detection interrupt.

VBTADCEn bits (VBATT Tamper Detection Backup Register Clear Enable) (n = 0, 1, 2)

The VBTADCEn (n = 0, 1, 2) bit enables backup register clearing by tamper detection flag.

12.2.16 VBTADCR2 : VBATT Tamper Detection Control Register 2

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC4A

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	VBRTCES2	VBRTCES1	VBRTCES0
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VBRTCES0	VBATT RTC Time Capture Event Source Select 0 0: RTCIC0 1: VBTADF0	R/W
1	VBRTCES1	VBATT RTC Time Capture Event Source Select 1 0: RTCIC1 1: VBTADF1	R/W
2	VBRTCES2	VBATT RTC Time Capture Event Source Select 2 0: RTCIC2 1: VBTADF2	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

The VBTADCR2 register control tamper detection functions. This register is only initialized by VBATT_POR reset.

VBRTCESn bits (VBATT RTC Time Capture Event Source Select) (n = 0, 1, 2)

The VBRTCESn (n = 0, 1, 2) bit selects RTC time capture event source.

When the noise cancellation function of the RTCIC pin input and the Sub-clock oscillation stop detection function are both enabled (VBTICTLR2.VCHnNCEN = 1 and SOSTDCR. SOSTDE = 1), set RTC Time Capture Event Source to VBTADFn (VBRTCESn = 1).

12.2.17 VBTADCR3 : VBATT Tamper Detection Control Register 3

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC54

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	VBTADZE2	VBTADZE1	VBTADZE0
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VBTADZE0	VBATT Tamper Detection Zeroization Enable 0 0: Zeroing HUK request by VBTADF0 flag is disable 1: Zeroing HUK request by VBTADF0 flag is enable	R/W
1	VBTADZE1	VBATT Tamper Detection Zeroization Enable 1 0: Zeroing HUK request by VBTADF1 flag is disable 1: Zeroing HUK request by VBTADF1 flag is enable	R/W
2	VBTADZE2	VBATT Tamper Detection Zeroization Enable 2 0: Zeroing HUK request by VBTADF2 flag is disable 1: Zeroing HUK request by VBTADF2 flag is enable	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

The VBTADCR3 register control tamper detection functions. This register is only initialized by VBATT_POR reset.

VBTADZEn bit (VBATT Tamper Detection Zeroization Enable) (n = 0, 1, 2)

The VBTADZEn (n = 0, 1, 2) bit enables HUK Zeroization requests output by tamper detection flag.

12.2.18 VBTNCWCR : VBATT Noise Canceler Width Control Register

Base address: SYSC = 0x4001_E000
SYSC_NS = 0x5001_E000

Offset address: 0xC50

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	VINCW[2:0]		
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	VINCW[2:0]	VBATT Input Noise Canceler Width select 0 0 0: 32.768 kHz 0 0 1: 64 Hz 0 1 0: 32 Hz 0 1 1: 16 Hz 1 0 0: 8 Hz 1 0 1: 4 Hz 1 1 0: 2 Hz 1 1 1: 1 Hz	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

This register is only initialized by VBATT_POR reset.

The VBTNCWCR bit selects noise cancellation width of Tamper input.

Denoising is done using the circuit shown in Figure 12.5. The setting is the frequency of the noise sampling clock.

When VBTNCWCR is set to a value other than 3'b000, start the RTC 64 Hz counter.

12.3 Operation

12.3.1 Battery Backup Function

When the voltage on the VCC pin drops, power can be supplied to the VBATT_R backup power area from the VBATT pin. The power supply from the VCC pin is resumed when the voltage on the VCC pin exceeds $V_{DETBATT}$. When a drop of power supply from VCC pin is detected, backup power area enter the VBATT Mode. In the VBATT Mode, power supply is switched to VBATT pin. The power supply from the VCC pin is resumed when the voltage on the VCC pin exceeds $V_{DETBATT}$. This power supply change does not affect the VBATT_R backup power area function.

It is necessary to enable voltage monitor 0 reset to use the battery backup

The VBATT_R backup power area include following functions:

- RTC (including time capture detection, triggered by a change of the time capture pin input level)
- Sub-clock oscillator (including XCIN and XCOUT pins)
- The sub-clock oscillation stopping detection function
- VBATT Backup Register
- Tamper Detection Function
- VBATT Voltage Monitor Function

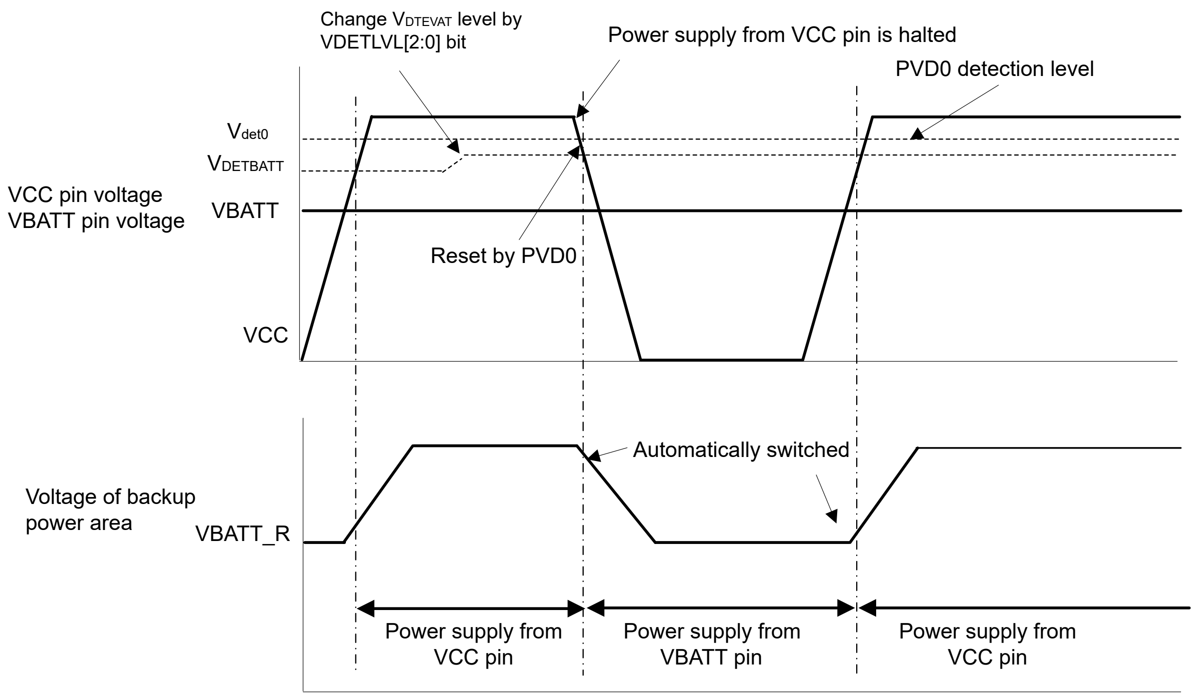
Table 12.1 shows the operating states in VBATT mode.

Table 12.1 Operating States in VBATT Mode

Operating state	VBATT Mode
Transition condition	Detection of VCC voltage drop
Canceling method other than reset	Detection of VCC voltage rise
State after cancellation by an interrupt	—
State after cancellation by a reset	—
Main clock oscillator	Stop
Sub-clock oscillator	Operating or not operating can be selected by SOSCCR.SOSTP bit. The status of the oscillator is same as before entering VBATT mode.
High-speed on-chip oscillator	Stop
Middle-speed on-chip oscillator	Stop
Low-speed on-chip oscillator	Stop
Sub-clock oscillation-stop detection function	Operating or not operating can be selected by SOSTDCR.SOSTDE bit. The status of the function is same as before entering VBATT mode.
PLL	Stop
PLL2	Stop
CPU	Stop (Undefined)
SRAM (ECC RAM included)	Stop (Undefined)
VBATT Backup Register	Stop (Retained or zeroized is selectable when Tamper input was detected.)
MRAM	Stop (Retained)
Realtime clock (RTC)	Selectable when selecting clock which is operating as the count source.
Programmable voltage detection circuit (PVD)	Stop
Power-on reset circuit	Stop
Other Peripheral modules	Stop (Undefined)
I/O ports	RTCICn ports (n = 0 to 2): Operating All ports not specified here: Undefined

Note: Stop (Retained) means that the contents of the internal registers are retained, but the operations are suspended.
Note: Stop (Undefined) means that the contents of the internal registers are undefined and power to the internal circuit is cut off.

Figure 12.2 shows switching sequence of Battery backup function.



Note: For details, see section 60, Electrical Characteristics.
Note: $V_{DETBATT}$ indicates the threshold level of the power supply change between the VCC pin and the VBATT pin.

Figure 12.2 Switching sequence of Battery backup function.

12.3.2 VBATT Battery Power Supply Switch Usage

The battery power supply switch can switch the power supply from the VCC pin to the VBATT pin when the voltage being applied to the VCC pin drops. When the voltage rises, this switch changes the power supply from the VBATT pin to the VCC pin.

The VCC voltage drop detection function for the battery power supply switch in "Figure 12.1" consists of two voltage drop detectors. One is a VCC drop detector for normal operation of the battery power supply switch that operates according to the settings of the VDTETLVL[2:0] bit and the VDETE bit. The other is voltage monitoring 0 to reduce power consumption during the Deep Software Standby mode 1 or 2.

In cold start, VDETE bit is initialized by VBATT_POR reset and the VCC voltage drop detection function for the battery power supply switch is disabled.

To use the battery power supply switch, the proper $V_{DETBATT}$ level must be selected by VDETLVL[2:0] bits and wait t_{DETWT} for stabilization. After wait, the VCC voltage drop detection function is enabled by VDETE bit. The battery power supply switch has a constraint that the $V_{DETBATT}$ level selected by VDETLVL[2:0] bits should be below Voltage monitoring 0 level.

When the MCU transitions to the Deep Software Standby mode 1 or 2 and the low power consumption function of voltage monitor 0 is enabled, the VCC drop detector for normal operation of the battery power supply switch is stopped to reduce power consumption. At this time, the power supply switch is controlled by the voltage monitor 0. This control by voltage monitor 0 continues until it returns from Deep Software Standby mode.

The battery backup function should be used after the voltage monitoring 0 reset is enabled (OFS1.PVDAS bit is 0).

If you don't use battery power supply switch. You must set BPWSWSTP bit to 1 and short the VCC and VBATT pins. When BPWSWSTP bit is 1, switch is stop, and power is always supplied from VCC pin.

12.3.3 VBATT_R Voltage Drop Detection Function Usage

This function enables to detect VBATT_R voltage drop. When VBATT_R voltage is drop and lower than $V_{PDR\ (BATR)}$, VBATT_POR reset is asserted. The VBPORF flag is set to 1 by VBATT_POR reset. If VBPORF flag is 1, RTC counter and register is invalid and Battery Backup Function registers are reset. You must initialize the functions included backup power area.

Figure 12.3 shows VBATT_R voltage drop detection function.

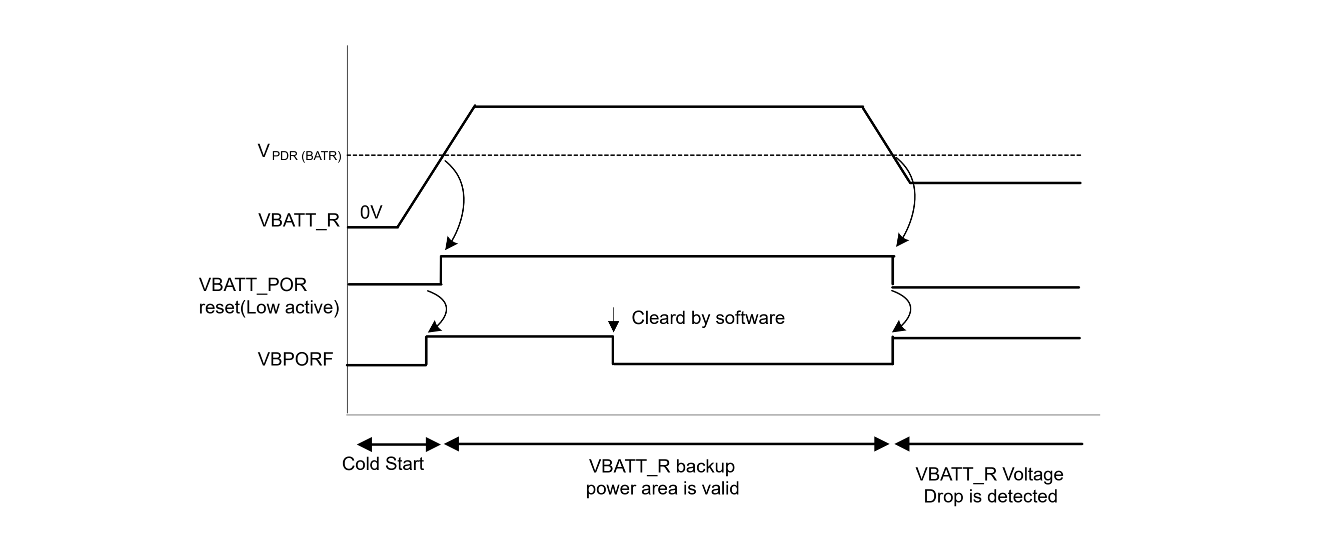


Figure 12.3 VBATT_R voltage drop detection function

12.3.4 VBATT Backup Register Usage

You can use 32, 16 or 8-bit access instruction when accessing the VBATT backup registers.

However, for example, when a 32-bit access instruction is executed, 8-bit read or write operation is executed with 4 consecutive times. When accessing, please note that the byte order of the data stored in the VBTBKR[n] (n = 0 to 127) register is little endian.

The data of VBATT backup register is cleared to 0x00 by VBATT_POR reset or tamper detection function.

The tamper detection function starts clearing operation of the VBATT backup register when the clear function is enabled and the tamper detection flag is set to 1. Do not cancel the clearing operation for 100 ns after starting. To cancel the clearing operation, disable the clear function or clear the tamper detection flag to 0. After canceling the clearing operation, do not access the VBATT backup register for 500 ns.

12.3.5 Tamper Detection Usage

The tamper detection function detects the RTCICn (n = 0 to 2) pin input event. The input event is defined as a change of RTCICn pin input level. The VBTADFn (n = 0 to 2) flag is set to 1 by the input event. When generate interrupt is enabled and the flag is set to 1, tamper detection interrupt is generated. When backup register clear function is enabled and flag is set to 1, the data of VBATT backup register is cleared. When Tamper Detection Zeroization Enable is enabled and flag is set to 1, the HUK Zeroization request is outputted. RTC time capture event source can select RTCICn pin input or the VBTADFn flag. If you need to synchronize time capture with a set of VBTADFn flags to 1, select the VBTADFn flag as the RTC time capture event source is recommended.

Figure 12.4 shows Tamper detection function.

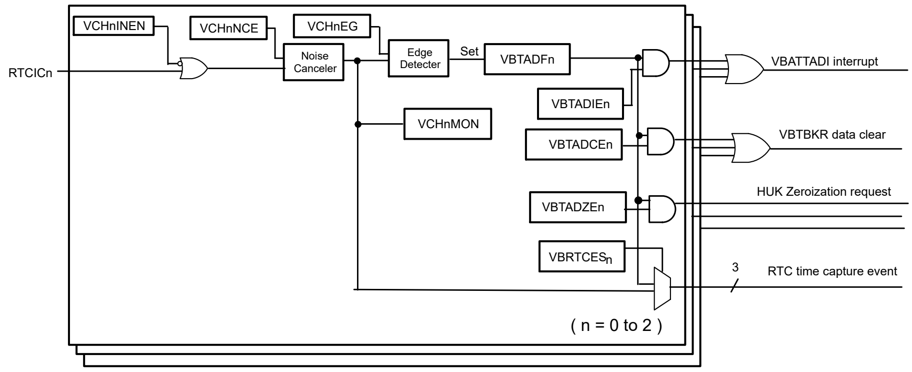


Figure 12.4 Tamper detection function

RTCICn pin inputs is enabled by the VCHnINEN (n = 0 to 2) bit. A wait time of 50us is required for the operation to stabilize after the RTCICn input is enabled.

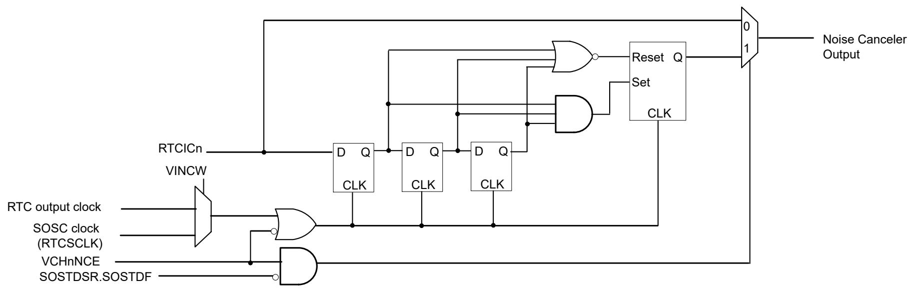
The VCHnNCE (n = 0 to 2) bit enables noise canceler. The noise canceler samples input signals at the Sub-Clock (RTCSCLK) or RTC output frequency and removes the pulses whose length is less than three sampling cycles.

The noise reduction sampling frequency can be selected in the VBTNCWCR.VINCW bit.

Figure 12.5 shows noise canceler circuit block diagram.

When changing the noise cancellation sampling frequency (changing the setting value of VBTNCWCR.VINCW bit), the VCHnNCE bit, VBTADCR1, VBTADCR2, and VBTADCR3 should be disabled.

The recommended initialization procedure is according to the flow example 3. For details, see section 12.3.7.4. Tamper Detection Function Initialization Setting Flow Example.



Note: n = 0 to 2

Figure 12.5 Noise canceler circuit block diagram

After enabling this noise canceler, a stabilization time of 5 clocks is required.

The VCHnEG (n = 0 to 2) bit enables to select Tamper detection edge. The edge detector operates asynchronously. When these control registers are changed, the VBTADFn flag may be set to 1 in a pseudo. The VBTADFn flag needs to be checked and clear to 0 after initialization of control registers. The VCHNnMON (n = 0 to 2) bit enables to monitor current input status. This bit also needs to be checked for inactive level after initialization of control registers.

The VBATTADI interrupt is asserted when 1 or more of 3 channels are flagged and interrupt is enabled.

The VBTBKRm(m = 0 to 127) register data is cleared to 00h when 1 or more of 3 channels are flagged and the VBATT backup register clear function is enabled.

12.3.6 VBATT Voltage Monitor Function Usage

You can monitor the input voltage level of VBATT pin. 1/6 of VBATT pin voltage level can be monitored as analog signal. After setting the VBTMNSEL bit to 1, it is necessary to wait t_{MONWT} for the monitor level to stabilize. For details on t_{MONWT}, see section 60, Electrical Characteristics.

Figure 12.6 shows VBATT voltage monitor function.

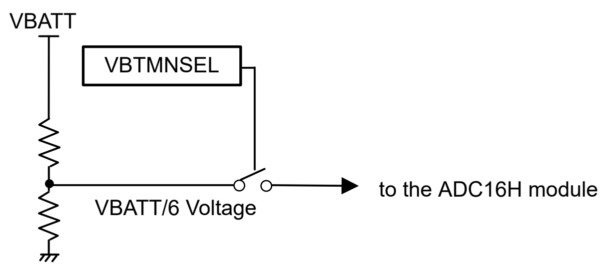


Figure 12.6 VBATT voltage monitor function

Note: For more information, see section 52, 16-bit A/D Converter (ADC16H).

Note: When VBTMNSEL bit is 1, VBAT power consumption is increased. It is recommended that VBTMNSEL bit sets to 1 only monitor timing.

12.3.7 Initial Settings Examples

12.3.7.1 Cold Start and Using the Power Supply Switch Flow Example

When turning on VCC pin and VBATT pin power for the first time, use the following flow to initialize.

1. Check VBPORM bit. If VBPORM flag is 0, wait until it changes to 1.
2. Clear VBPORF flag to 0.
3. Set the VDETLVL[2:0] bit to appropriate value. The $V_{DETBATT}$ level is Selected.
4. Wait t_{DETWT} for the VCC voltage detection function stabilization.
5. Set the VDETE bit to 1. The VCC voltage detection function is enabled.
6. Enable Sub-Clock Oscillator if needed.
7. Set Other Battery Backup Function and RTC registers.

For details on t_{DETWT}, see section 60, Electrical Characteristics.

For details on Sub-Clock Oscillator, see section 9, Clock Generation Circuit.

12.3.7.2 Warm Start and Using the Power Supply Switch Flow Example

The MCU transitions to VBATT mode when the power of the VCC pin is turned off while the power of the VBATT pin is on. In this section, warm start means when the power supply of the VCC pin is turned on in VBATT mode. When the MCU starts processing with a warm start, execute the following flow in order to check the VBATT_R voltage drop before executing other process.

1. Check VBPORM flag. If VBPORM flag is 0, wait until it changes to 1.
2. Check VBPORF flag. If VBPORF flag is 1, the VBATT_R voltage drop is detected. For details on re-initialize the backup power area, see section 12.3.7.1. Cold Start and Using the Power Supply Switch Flow Example. If not, it means that the VBATT_R voltage does not drop and the state of the backup power area is retained. So, you do not need to re-initialize the backup power area.

12.3.7.3 Not Using the Power Supply Switch Flow Example

When not using the power supply switch, VCC pin and VBATT pin should be shorted. In this case, the VBATT_POR reset cannot follow the power on reset of VCC and VBATT_R voltage drop may not be detected. So, initialize the backup power supply area with the following flow.

1. Set the BPWSWSTP bit to 1. The power supply switch is stopped.
2. Check VBPORM flag. If VBPORM flag is 1, wait until it changes to 0.
3. Clear the VDETE bit to 0. The VCC voltage drop detection function is stopped.
4. Clear the VDETLVL[2:0] bit to 110b. The initial value is selected.
5. Check VBPORF flag. When VBPORF flag is 1, clear it to 0.
6. Set the SOSTP bit to 1 regardless of its value. Stop Sub-Clock Oscillator.
7. Initialize the VBTICTLR register and SOMCR.SOSEL bit. Initialization is recommended because these registers are related to the control of the IO port.
8. Initialize Other Battery Backup Function registers if needed. Other Battery Backup Function registers are VBTICTLR2, VBTADSR, VBTADCR1, VBTADCR2, and VBTBKRn.
9. Enable Sub-Clock Oscillator if needed.
10. Set RTC registers.

For details on Sub-Clock Oscillator, see section 9, Clock Generation Circuit.

12.3.7.4 Tamper Detection Function Initialization Setting Flow Example

If using the tamper detection function, it is recommended to initialize according to the following flow example.

1. Set the VCHnINEN to appropriate value.
2. Wait 50us to stabilize after the RTCICn input.
3. Set the VBTNCWCR to appropriate value.
4. Set the VCHnNCE bit and the VCHnEG bit.
5. Wait 5 RTC-clocks to stabilize, if noise canceler is enabled.
6. Checking the VCHnMON bit indicates inactive is recommended. If it is active, it may not be possible to detect the tamper.
7. Initialize the invalid status of VBTADFn flag by clear to 0 after reading dummy read.
8. Set VBTADCR1, VBTADCR2,VBTADCR3 to enable interrupt, backup register clear, HUK Zeroization request output, and RTC time capture event.
9. Enables RTC time capture function if needed.

12.4 Interrupt Sources

The Battery Backup Function has two interrupt sources and are listed in Table 12.2.

Table 12.2 Battery Backup Function interrupt Sources

Symbol	Interrupt source	Interrupt flag	Interrupt conditions
VBATTADI	VBATT Tamper detection	VBTADF0	VBTADF0 = 1, VBTADIE0 = 1
		VBTADF1	VBTADF1 = 1, VBTADIE1 = 1
		VBTADF2	VBTADF2 = 1, VBTADIE2 = 1

12.5 Usage Notes

1. Operation of the sub-clock oscillator and RTC are not guaranteed when the voltage level on VBATT is lower than the guaranteed operation range. Initialize the RTC when the VBATT pin falls below the guaranteed operating voltage and then powers up again.
2. A reset generated while writing to registers described in this section might destroy the register value.

3. When VCC is higher than $V_{DETBATT\_m}$, the VCC pin and VBATT pin are separated. When VCC is lower than $V_{DETBATT}$ and the switch is connected to the VBATT pin, and if the voltage on VBATT drops lower than VCC, current might flow into the VBATT pin through the parasitic diode between the VCC and VBATT pins.
4. During RTC operation using the voltage from the VBATT pin and the I/O ports within the backup, the power supply area can only be used as time capture event input pins for the RTC.